Data Base System

24k-0705 | Muhammad Adil Saeed

In Lab:

1.

Worksheet | Query Builder

```
SELECT SUM(salary) AS total_salary
FROM employees;
```

Query Result

SQL | All Rows Fetched: 1 in 0.004 seconds

	TOTAL_SALARY
1	691416

2.

Worksheet | Query Builder

```
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id;
```

Query Result

SQL | All Rows Fetched: 12 in 0.003 seconds

	DEPARTMENT_ID	AVG_SALARY
1	100	8601.3333333333333333333333333333333333
2	30	4150
3	(null)	7000
4	90	19333.3333333333333333333333333333333333
5	20	9500
6	70	10000
7	110	10154
8	50	3475.5555555555555555555555555555555556
9	80	8955.8823529411764705882352941176470588 24
10	40	6500
11	60	5760
12	10	4400

3.

Worksheet | Query Builder

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id;
```

Query Result

SQL | All Rows Fetched: 12 in 0.003 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEES
1	100	6
2	30	6
3	(null)	1
4	90	3
5	20	2
6	70	1
7	110	2
8	50	45
9	80	34
10	40	1
11	60	5
12	10	1

4.

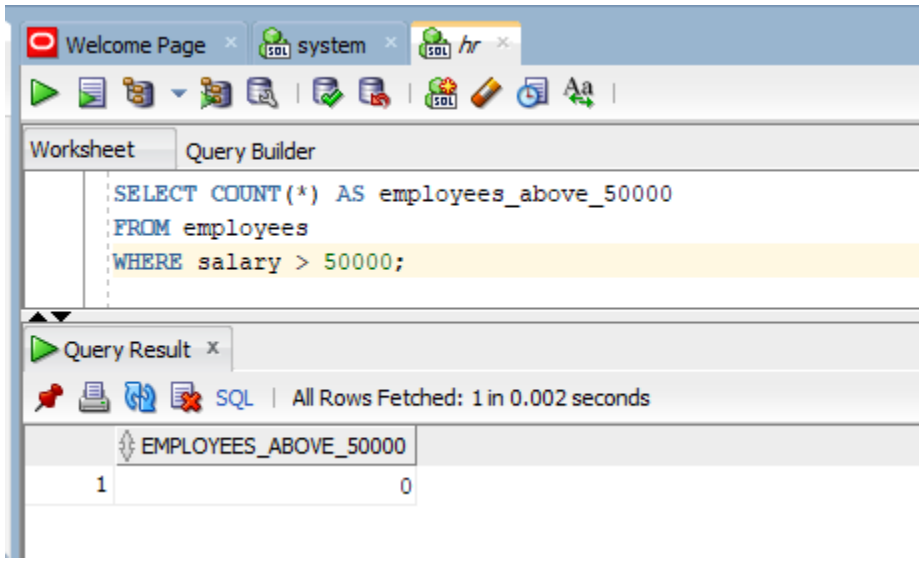
```
SELECT job_id,
       MIN(salary) AS min_salary,
       MAX(salary) AS max_salary
FROM employees
GROUP BY job_id;
```

Script Output | Query Result

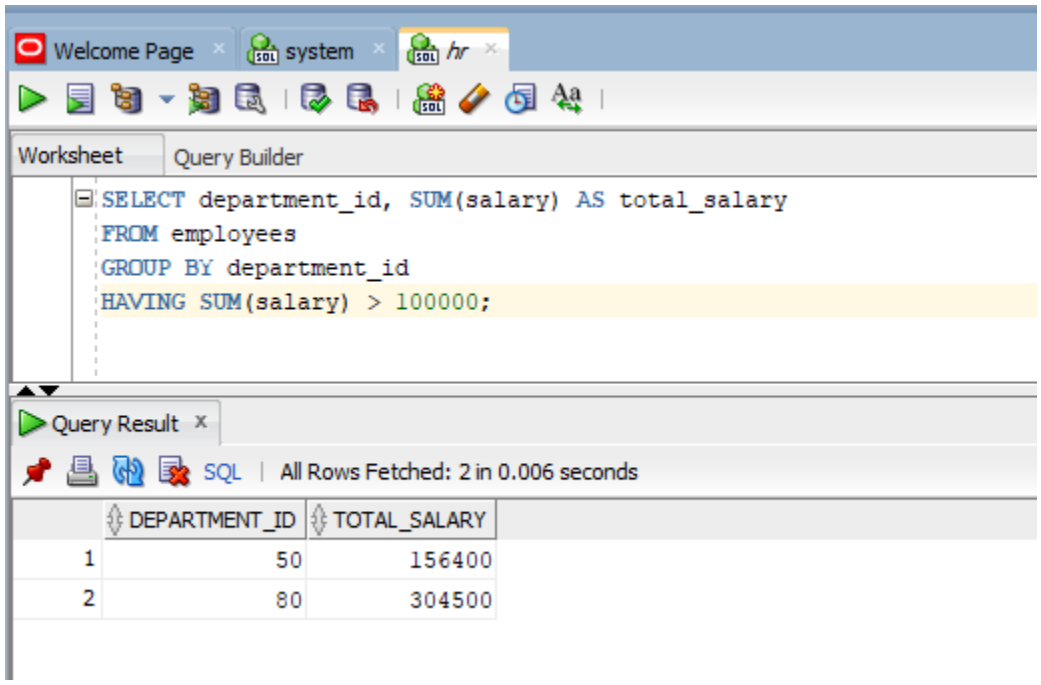
SQL | All Rows Fetched: 19 in 0.003 seconds

	JOB_ID	MIN_SALARY	MAX_SALARY
1	IT_PROG	4200	9000
2	AC_MGR	12008	12008
3	AC_ACCOUNT	8300	8300
4	ST_MAN	5800	8200
5	PU_MAN	11000	11000
6	AD_ASST	4400	4400
7	AD_VP	17000	17000
8	SH_CLERK	2500	4200
9	FI_ACCOUNT	6900	9000
10	FI_MGR	12008	12008
11	PU_CLERK	2500	3100
12	SA_MAN	10500	14000
13	MK_MAN	13000	13000
14	PR_REP	10000	10000
15	AD_PRES	24000	24000
16	SA_REP	6100	11500
17	MK_REP	6000	6000
18	ST_CLERK	2100	3600
19	HR_REP	6500	6500

5.


Welcome Page
system
hr
Worksheet
Query Builder
SELECT COUNT(*) AS employees_above_50000
FROM employees
WHERE salary > 50000;
Query Result
SQL
All Rows Fetched: 1 in 0.002 seconds
EMPLOYEES_ABOVE_50000
1
0

6.


Welcome Page
system
hr
Worksheet
Query Builder
SELECT department_id, SUM(salary) AS total_salary
FROM employees
GROUP BY department_id
HAVING SUM(salary) > 100000;
Query Result
SQL
All Rows Fetched: 2 in 0.006 seconds
DEPARTMENT_ID
TOTAL_SALARY
1
50
156400
2
80
304500

7.

```sql
SELECT d.department_name, COUNT(e.employee_id) AS total_employees
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
GROUP BY d.department_name;
```

Script Output | Query Result

SQL | All Rows Fetched: 11 in 0.006 seconds

	DEPARTMENT_NAME	TOTAL_EMPLOYEES
1	Administration	1
2	Accounting	2
3	Purchasing	6
4	Human Resources	1
5	IT	5
6	Public Relations	1
7	Executive	3
8	Shipping	45
9	Sales	34
10	Finance	6
11	Marketing	2

8.

Welcome Page | system | hr

Worksheet | Query Builder

```sql
SELECT department_id, SUM(salary) AS total_salary
FROM employees
GROUP BY department_id
HAVING SUM(salary) > 150000;
```

Query Result

SQL | All Rows Fetched: 2 in 0.002 seconds

	DEPARTMENT_ID	TOTAL_SALARY
1	50	156400
2	80	304500

9.

```
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE d.department_name LIKE 'S%'
GROUP BY d.department_name;
```

All Rows Fetched: 2 in 0.006 seconds

	DEPARTMENT_NAME	AVG_SALARY
1	Shipping	3475.5555555555555555555555555555555556
2	Sales	8955.8823529411764705882352941176470588 24

10.

```
INSERT INTO employees
(EMPLOYEE_ID, FIRST_NAME, LAST_NAME, EMAIL, HIRE_DATE, JOB_ID, SALARY, DEPARTMENT_ID)
VALUES
(207, 'Ali', 'Saeed', 'ASAEED207', SYSDATE, 'IT_PROG', 65000, 60);
```

Task completed in 0.02 seconds

```
1 row inserted.
```

Post Lab:

11.

```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5;
```

All Rows Fetched: 5 in 0.002 seconds

	DEPARTMENT_ID	TOTAL_EMPLOYEES
1	100	6
2	30	6
3	50	45
4	80	34
5	60	6

12.

```sql
SELECT department_id, AVG(salary) AS avg_salary
FROM employees
GROUP BY department_id
HAVING AVG(salary) > 60000;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 0 in 0.003 seconds

DEPARTM...	AVG_...

13.

```sql
SELECT TO_CHAR(hire_date, 'YYYY') AS hire_year,
       COUNT(*)
FROM employees
GROUP BY TO_CHAR(hire_date, 'YYYY')
ORDER BY hire_year;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 9 in 0.001 seconds

	HIRE_YEAR	COUNT(*)
1	2001	1
2	2002	7
3	2003	6
4	2004	10
5	2005	29
6	2006	24
7	2007	19
8	2008	11
9	2026	1

14.

```sql
SELECT d.department_name, COUNT(e.employee_id) AS total_employees
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY d.department_name;
```

Script Output × Query Result ×

SQL | All Rows Fetched: 11 in 0.003 seconds

	DEPARTMENT_NAME	TOTAL_EMPLOYEES
1	Accounting	2
2	Administration	1
3	Executive	3
4	Finance	6
5	Human Resources	1
6	IT	6
7	Marketing	2
8	Public Relations	1
9	Purchasing	6
10	Sales	34
11	Shipping	45

15.

```
SELECT job_id, COUNT(*) AS total_employees
FROM employees
GROUP BY job_id
HAVING COUNT(*) > 3;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 8 in 0.002 seconds

	JOB_ID	TOTAL_EMPLOYEES
1	FI_ACCOUNT	5
2	IT_PROG	6
3	PU_CLERK	5
4	SA_MAN	5
5	SA_REP	30
6	SH_CLERK	20
7	ST_CLERK	20
8	ST_MAN	5

16.

```
SELECT department_id, MAX(salary) AS max_salary
FROM employees
GROUP BY department_id
HAVING MAX(salary) > 80000;
```

Script Output × | Query Result ×

SQL | All Rows Fetched: 0 in 0.002 seconds

DEPARTM...	MAX_SAL...

17.

```
INSERT ALL
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (271, 'Sales_New', NULL, 1700)
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (272, 'Marketing_New', NULL, 1700)
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (273, 'HR_New', NULL, 1700)
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (274, 'Finance_New', NULL, 1700)
  INTO departments (department_id, department_name, manager_id, location_id)
    VALUES (275, 'IT_Support_New', NULL, 1700)
SELECT * FROM dual;
```

Script Output × | Query Result ×

Task completed in 0.019 seconds

```
5 rows inserted.
```

18.

Worksheet | Query Builder

```
CREATE TABLE tasks AS
SELECT employee_id, first_name, job_id
FROM employees
WHERE 1=2;
```

Script Output | Query Result

Task completed in 0.022 seconds

```
Table TASKS created.
```

```
INSERT INTO tasks (employee_id, first_name, job_id)
SELECT employee_id, first_name, job_id
FROM employees;
```

Script Output | Query Result

Task completed in 0.023 seconds

```
Table TASKS created.


108 rows inserted.
```

19.

Worksheet | Query Builder

```
UPDATE employees
SET department_id = 60
WHERE employee_id = 112;
```

20.

Worksheet | Query Builder

```
DELETE FROM job_history
WHERE employee_id IN (201, 202);

DELETE FROM employees
WHERE employee_id IN (201, 202);

commit;
```

Script Output | Query Result

Task completed in 0.024 seconds

```
0 rows updated.

1 row deleted.

2 rows deleted.

Commit complete.
```